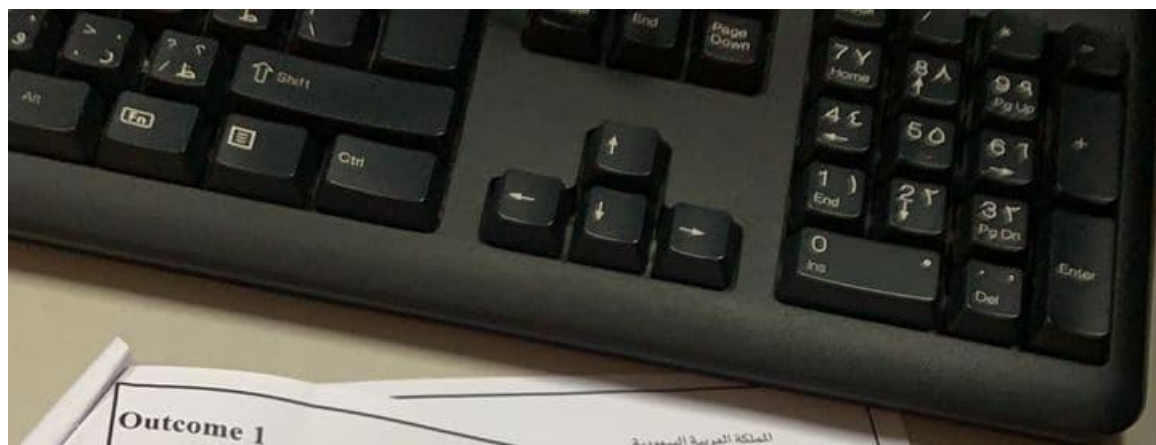




Outcome 1

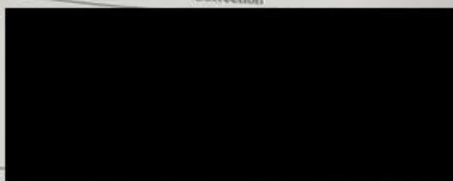
المملكة العربية السعودية
وزارة التعلّم

- 1.1. An ability to analyze a complex computing problem, and identify the principles of computing and other relevant disciplines appropriate to its solution. (1)
- 1.2. An ability to solve a complex computing problem by applying principles of computing and other relevant disciplines appropriate to its solution. (1)
1. Examine the following program then identify the errors: [25 Marks]

Program

```
public class COCS202 {  
    public static void main(String[] args) {  
        String [3] m = {"KHOJ", "FCITR", "KAU"};  
  
        do {  
            System.out.println(m);  
            i = i + 1;  
        } while(i < m.length)  
    } //end main  
} //end class
```

Correction



2. Analyze the following code fragment then define the output: [1,1,8]

#	Code fragment	Output
1	<pre>int x = 6 % 2; int y = 6 / 2; System.out.println(x + "n" + y);</pre>	
2	<pre>int x = 0; do { ++x; System.out.println(x); } while(x < 2);</pre>	
3	<pre>public class COCS202 { public static void abc(String m, int n) { for (int i = 1; i <= n; i++) System.out.println(m); } public static void main(String args[]) { abc("KAU", 2); } //end main } //end class</pre>	
4	<pre>int[] values = new int[2]; for (int i = 0; i < values.length; i++) { values[i] = 2 * i + 1; System.out.println(values[i]); }</pre>	

3. Analyze the following code fragment then fill the gaps to run the following output: [1,1,8]

Programs	Output
<pre>for (int i =; i <=; i++) { for (int j =; j <=; j++) { if (..... % 2 == 1) System.out.print(..... + "\t"); else System.out.print(..... + "\t"); } System.out.print(.....); }</pre>	<pre>1 2 3 4 5 * * * * * 1 2 3 4 5 * * * * *</pre>

4. Write the correct answer to the following two-dimensional array.

(1.2: 5)

```
int [ ][ ] list = { {1, 2}, {3, 4, 5}, {6, 7, 8, 9, 10} };
```

- What is the length of list.length?
- What is the length of list [2].length?
- What is the length of list [3].length?
- What is the value of list [1][1]?
- What does the following code segment print?
for (int j = 0; j < 2; j++)
System.out.print(list [0][j] + " ");

[15 Marks]

Outcome 2

- 2.1: An ability to design a computer-based solution to meet a given set of computing requirements in the context of the program's discipline. (1)
- 2.2: An ability to implement a computer-based solution to meet a given set of computing requirements in the context of the program's discipline. (1)

(2.1: 5)

5. Design a flowchart that read number from user then it will check whether a number is positive or negative.

6. Write a Java program that meets the following requirements

- Define the getAverage method:
 - a. You have to compute average using for loop. (4 marks)
 - b. Return average to the main. (1 mark)
- Main method:
 - c. Create an array with the following data 50, 10 and 3 (Declaring, creating and initializing in one step) (1 mark)
 - d. Invoke getAverage method. (1 mark)
 - e. Print how many numbers in the array above the average. (using for loop and if statement) (3 marks)

```
public class COCS202 {  
    public static double getAverage(int[] array) {
```

```
    } //end getAverage
```

```
    public static void main(String args[]) {
```

```
    } //end main  
}
```